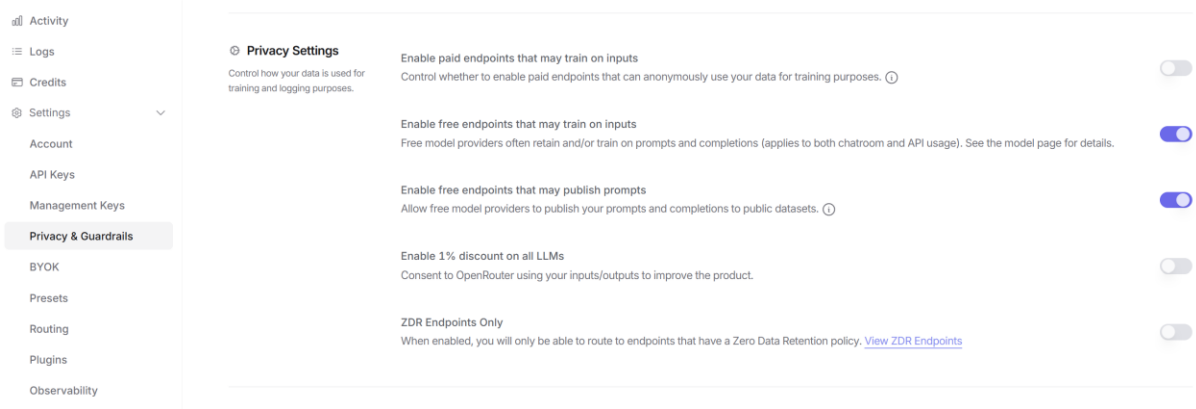
In the “Thinking_Models_Task.ipynb” notebook, the training dataset generation part is from the start of the notebook to before the “Configure Unsloth” section, and the rest is the model fine-tuning part. The “.env” template and the JSON generated datasets are included in the return ZIP file. If you use Google Colab, please remember to upload these files (except “.env” file) to session storage. Moreover, the “Thinking_Models_Task.py” is the Python script version exported from the original “Thinking_Models_Task.ipynb” notebook.

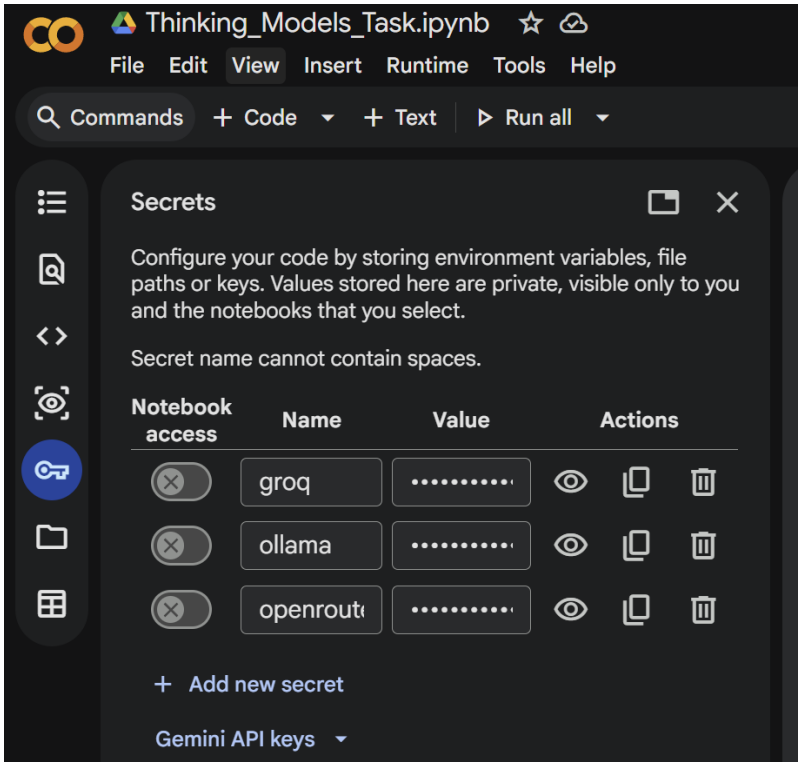
I recommend using Google Colab with T4 (or better) GPU for faster training. For local machine, I used Windows 11 with Nvidia CUDA and Python 3.12 to run the code.

Before running the code, please do a few things below:

- Create an account/Login and create the API keys on these websites (all of them are used in the dataset generation part).:
    - Groq: https://console.groq.com/keys
    - Ollama: https://ollama.com/settings/keys **AND/OR** OpenRouter: https://openrouter.ai/settings/keys (I recommend using Ollama only)
- On OpenRouter, after creating API keys, go to Settings > Privacy & Guardrails > Privacy Settings, then enable the second and third options (as shown in the below screenshot). This will allow gpt-oss free models to run programmatically via API calls.



- If you use local machine, please fill in the API keys in the provided “.env” template file. If you use Google Colab, add the keys in “Secrets” (the key button) on the left side of the window (as shown in the below screenshot).



Then, remember to run the first few cells to install required packages. Now the notebook is ready to run. Other instruction and detailed descriptions about the code can be read and follow throughout the notebook.

While installing packages & loading API keys, be aware if you use local machine or Google Colab (There are some comment lines that write “# For local machine” and “# For Google Colab”). Then, choose the correct lines to run and comment the other ones.

On Google Colab, while installing packages, if a dialogue appears and restart session is required, just restart it. You should rerun the installation to ensure all necessary packages have already been installed.